copying-fidelity. We then simply define a ‘gene’ as the largest entity which, at least potentially, has these properties. The gene is a long-lived replicator, existing in the form of many duplicate copies. It is not infinitely long-lived. Even a diamond is not literally everlasting, and even a cistron can be cut in two by crossing-over. The gene is defined as a piece of chromosome which is sufficiently short for it to last, potentially, for *long enough* for it to function as a significant unit of natural selection.

Exactly how long is ‘long enough’? There is no hard and fast answer. It will depend on how severe the natural selection ‘pressure’ is. That is, on how much more likely a ‘bad’ genetic unit is to die than its ‘good’ allele. This is a matter of quantitative detail which will vary from example to example. The largest practical unit of natural selection—the gene—will usually be found to lie somewhere on the scale between cistron and chromosome.

It is its potential immortality that makes a gene a good candidate as the basic unit of natural selection. But now the time has come to stress the word ‘potential’. A gene *can* live for a million years, but many new genes do not even make it past their first generation. The few new ones that succeed do so partly because they are lucky, but mainly because they have what it takes, and that means they are good at making survival machines. They have an effect on the embryonic development of each successive body in which they find themselves, such that that body is a little bit more likely to live and reproduce than it would have been under the influence of the rival gene or allele. For example, a ‘good’ gene might ensure its survival by tending to endow the successive bodies in which it finds itself with long legs, which help those bodies to escape from predators. This is a particular example, not a universal one. Long legs, after all, are not always an asset. To a mole they would be a handicap. Rather than bog ourselves down in details, can we think of any *universal* qualities that we would expect to find in all good (i.e. long-lived) genes? Conversely, what are the properties that instantly mark a gene out as a ‘bad’, short-lived one? There might be several such universal properties, but there is one that is particularly relevant to this book: at the gene level, altruism must be bad and selfishness good. This follows inexorably from our definitions of altruism and selfishness. Genes are competing directly with their alleles for survival, since their alleles in the gene pool are rivals for their slot on the chromosomes of